✅ THEORETICAL ANSWERS

Q1. Calculated Field

A calculated field is a custom formula created in Tableau using existing dataset fields.
Importance:

- Helps create new insights (Profit Ratio, Delivery Status)
- Enables dynamic analysis
- Used for KPIs and dashboards

Q2. Why parameters don't change visuals directly?

Parameters are **independent values**.
They only affect visuals when used inside:

- Calculated fields
- Filters
- Reference lines

Q3. When to use EXCLUDE LOD?

Use EXCLUDE when:

- You want to **remove a dimension from view level**
- Example: Showing category-wise data but ignoring product-level detail

---

Q4. FIXED vs INCLUDE

FIXED	INCLUDE
Ignores filters (except context)	Works with view level
Fixed level of detail	Adds extra detail
Example: City avg sales	Product-level avg inside category

✅ PRACTICAL TASKS (STEP-BY-STEP)

🔹 **Task 1: Delivery Performance Classification**

👉 **Create Calculated Field:**

Name: Delivery Status

```
IF [Delivery_Time_Min] <= 30 THEN "Fast"
ELSE "Delayed"
END
```

👉 **Use:**

- Drag to Color / Rows
- Count of Orders → see Fast vs Delayed

🔹 **Task 2: Loss-Making Orders**

👉 **Create Calculated Field:**

Name: Loss Flag

```
IF [Profit] < 0 THEN "Loss"
ELSE "Profit"
END
```

👉 **Use:**

- Add to Filter → select “Loss”
- Or use in chart (City vs Loss Orders)

🔹 **Task 3: Top N Products (Parameter)**

👉 **Step 1: Create Parameter**

Name: Top N

- Data Type: Integer
- Current Value: 5

👉 **Step 2: Create Calculated Field**

Name: Top N Filter

RANK(SUM([Sales])) <= [Top N]

👉 **Step 3:**

- Drag Product_Name → Rows
- SUM(Sales) → Columns
- Apply Top N Filter → TRUE

🔹 **Task 4: City-Level Benchmarking (FIXED LOD)**

👉 **Create Calculated Field:**

Name: City Avg Sales

{ FIXED [City] : AVG([Sales]) }

👉 **Use:**

- Compare:
  - Actual Sales vs City Avg Sales
- Add to Tooltip or dual axis chart

🔹 **Task 5: Product-Level Impact (INCLUDE LOD)**

👉 **Create Calculated Field:**

Name: Product Avg Sales

{ INCLUDE [Product_Name] : AVG([Sales]) }

👉 **Use:**

- Works even if Product is not in view
- Use at Category / City level

✅ **DASHBOARD IDEA (IMPORTANT FOR ASSIGNMENT)**

Create 1 dashboard with:

📊 **Charts:**

1. **Delivery Status Pie Chart**
2. **Loss Orders by City (Bar Chart)**
3. **Top N Products (Dynamic Bar Chart)**
4. **City Avg vs Actual Sales (Dual Axis)**
5. **Category-wise Product Avg Sales**

✅ **BONUS KPIs (Optional but impressive)**

Profit Ratio:

SUM([Profit]) / SUM([Sales])

Avg Delivery Time:

AVG([Delivery_Time_Min])